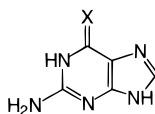


purines exhibited similar activity with regard to their affinity for the benzodiazepine receptor. In this example of bioisosteric replacement, pharmacological activity was retained. It is important to note that retention of biological activity based on *in vitro* data can be critical in those instances where differences between bioisosteric analogues exist with regard to *in vivo* parameters which may include absorption, distribution, metabolism, or elimination. While one may only observe retention of activity associated with interaction of drug with the pharmacophore, bioisosteres may differ dramatically in their *in vivo* efficacy. Additional examples of this bioisosteric replacement will be discussed in the next section on monovalent replacement of hydroxyl and thiol groups.

3. Interchange of Hydroxyl and Thiol Groups

The interchange of thiol for hydroxyl can be considered as an extension of the amino-hydroxyl replacement and has been used extensively in medicinal chemistry. This replacement is based on the ability of both these functional groups to be *hydrogen bond acceptors or donors*. A classical illustration of this replacement being guanine (**10a**) and 6-thioguanine (**10b**, Figure 7).²¹



10a X = O Guanine

10b X = S 6-Thioguanine

Figure 7.

As discussed in the previous section, when part of a heteroaromatic ring, these functional groups can exist in different tautomeric forms. Figure 8 illustrates the most common example wherein a mobile proton on a nitrogen atom in the aromatic ring can be transferred to the heteroatom attached to the adjacent carbon resulting in the different tautomers.

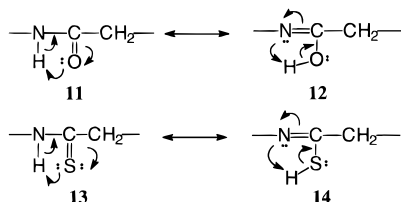


Figure 8.

In the case of 6-thioguanine, the ability of this bioisosteric analogue to be viewed as a substrate by the salvage pathway associated with purine biosynthesis, allows for its transformation into 6-thioguananylic acid by hypoxanthine-guanine phosphoribosyl-transferase (HGPRT). However, the significance of this "fraudulent" nucleic acid with respect to its lethality to neoplasms is uncertain.²² It is as this phosphoriboside that either the *de novo* synthesis of nucleic acids is inhibited or incorporation into deoxyribonucleic acid occurs.

In an attempt to enhance the calcium channel blocking capacity of certain dihydropyrimidine agents, a number of isosteric analogues with the general structure **15** (Figure 9) were synthesized.²³ Substitution of the hydroxyl with an amino resulted in

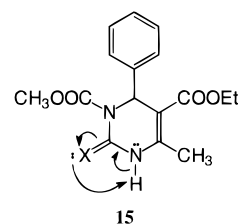


Figure 9.

Table 7. Calcium Channel Blocking Activity of 1,4-Dihydropyrimidines

compound	X	van der Waal's radius ²⁴ (Å)	IC ₅₀ (nM) ^a
15a	=O	1.40	140
15b	=NH	1.50	160
15c	=S	1.85	17

^a Concentration that produced 50% inhibition and determined for the vasorelaxant activity with potassium-depolarized rabbit thoracic aorta.

analogues with similar potency. However, substitution with the thiol resulted in enhanced potency (Table 7). This could be explained by the fact that the size of the substituents, described here as the van der Waal's radii, and the ability to hydrogen bond were the important factors influencing retention of activity. Therefore, replacement with the amino group, which has a similar size, resulted in similar potency. However, replacement with the sterically optimal thiol resulted in an analogue which was an order of magnitude more potent.

The use of this replacement in the design of novel anti-inflammatory agents substantiates its utility as a monovalent bioisostere. Long term use of nonsteroidal anti-inflammatory drugs (NSAIDs) for the treatment of rheumatoid arthritis and other inflammatory diseases has been associated with side effects such as gastrointestinal ulceration, bleeding, and nephrotoxicity.^{25,26} With a view to designing new drugs with an improved safety profile, certain thiazoles (**16**, Figure 10 and Table 8) that are dual

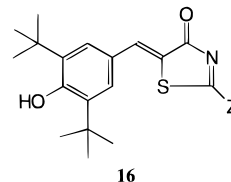


Figure 10.

Table 8. Anti-inflammatory Activity of Benzylidene Derivatives in Intact Rat Basophilic Leukemia (RBL-1) Cells

compound	Z	electronegativity ²⁹	IC ₅₀ (μM) ^a	
			5-LO	CO
16a	OH	3.51	1.4	0.35
16b	NH ₂	2.61	0.77	0.39
16c	SH	2.32	0.38	0.012

^a Concentration of the test compound causing 50% inhibition of 5-LO or CO formation.

inhibitors of both cyclooxygenase (CO) and 5-lipoxygenase (5-LO) are being studied as potential anti-inflammatory agents.²⁷ The beneficial effects of NSAIDs have been attributed to the inhibition of the enzyme cyclooxygenase, thereby preventing production of pro-inflammatory prostaglandins.²⁸ Leuko-